

The Von Neumann Architecture

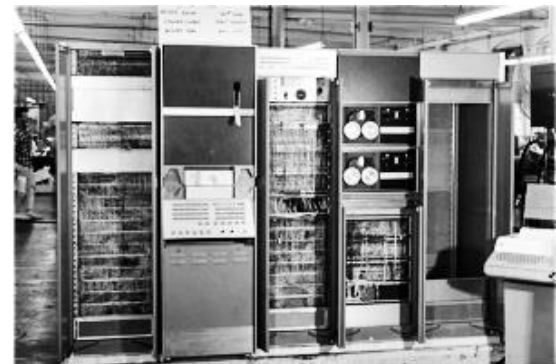


**Von Neumann
Architecture**



Designing Computers

- All computers more or less based on the same basic design, the Von Neumann Architecture!



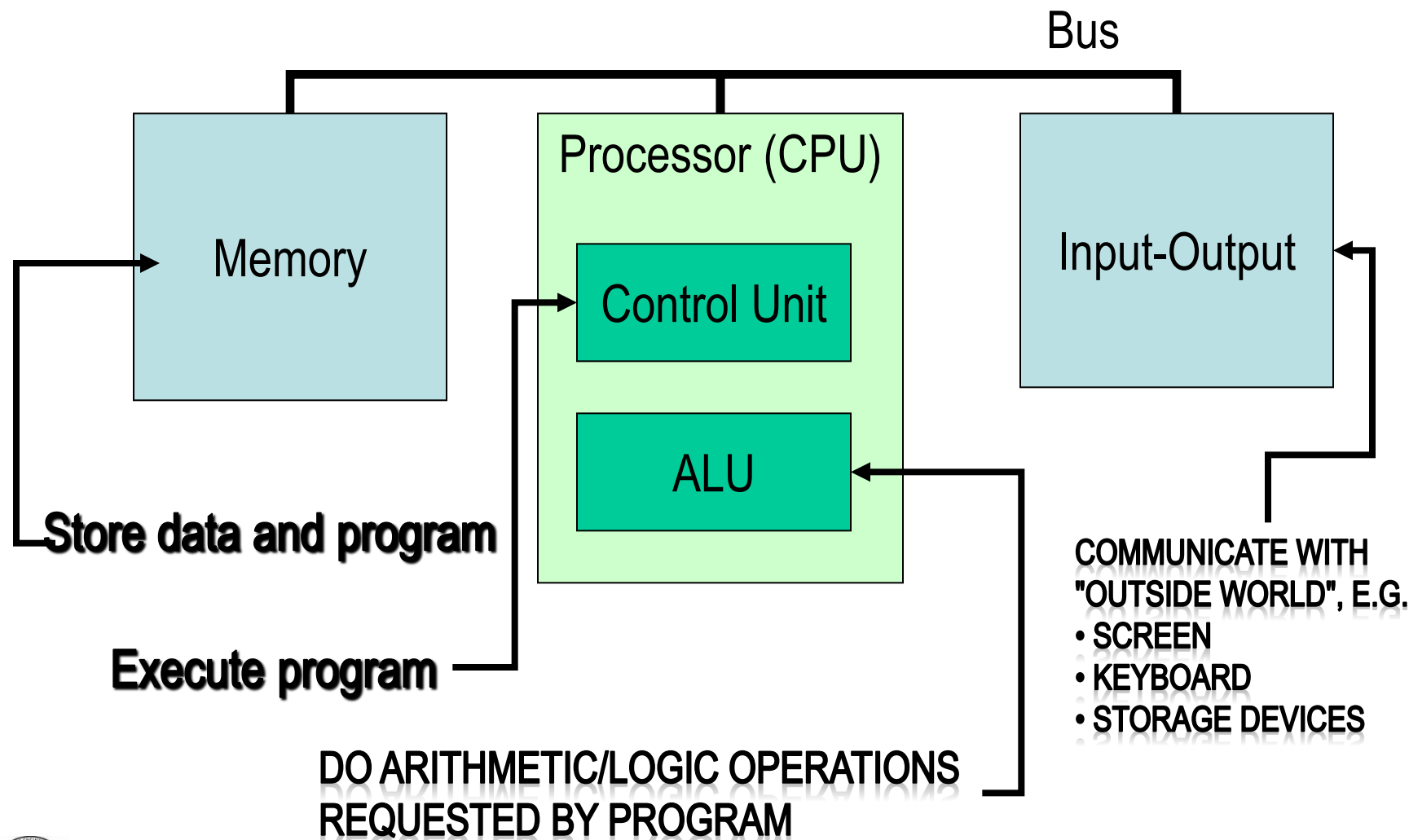
The Von Neumann Architecture

Model for designing and building computers, based on the following three characteristics:

- 1) The computer consists of four main sub-systems:
 - Memory
 - ALU (Arithmetic/Logic Unit)
 - Control Unit
 - Input/output System (I/O)
- 2) Program is stored in memory during execution.
- 3) Program instructions are executed sequentially.



The Von Neumann Architecture



Memory Subsystem

- Memory, also called RAM (Random Access Memory),
 - Consists of many memory cells (storage units) of a fixed size. Each cell has an address associated with it.
 - All accesses to memory are to a specified address. A cell is the minimum unit of access (fetch/store a complete cell).
 - The time it takes to fetch/store a cell is the same for all cells.
- When the computer is running, both
 - Program
 - Data (variables)are stored in the memory (same).



RAM

- Need to distinguish between
 - the address of a memory cell
 - and the content of a memory cell
- Memory width (W):
 - How many bits is each memory cell, typically one byte (=8 bits)
- Address width (N):
 - How many bits used to represent each address, determines the maximum memory size = address space
 - If address width is N -bits, then address space is 2^N ($0, 1, \dots, 2^N - 1$)

